

ABOUT THE TEST FoundationOne®Liquid CDx is a next generation sequencing (NGS) assay that identifies clinically relevant genomic alterations in circulating cell-free DNA.

PATIENT
DISEASE Stomach GIST
NAME 03-2026-00152703, UA
DATE OF BIRTH 25 April 1976
SEX Female
MEDICAL RECORD # CS5220323

PHYSICIAN
ORDERING PHYSICIAN HCP, Mol outsource
MEDICAL FACILITY MEDICHA LABORATORIA CSD
ADDITIONAL RECIPIENT None
MEDICAL FACILITY ID 324370
PATHOLOGIST Not Provided

SPECIMEN
SPECIMEN ID 00201593 1976/04/25
SPECIMEN TYPE Blood
SPECIMEN COLLECTION DATE 20 August 2026
SPECIMEN RECEIVED 25 August 2026

Biomarker Findings

Blood Tumor Mutational Burden - 3 Muts/Mb
ctDNA Tumor Fraction - High (32%)
Microsatellite status - MSI-High Not Detected

Genomic Findings

For a complete list of the genes assayed, please refer to the Appendix.

KIT exon 11 deletion (Q556_V559>H), exon 13 (V654A), exon 18 (A829P)

BRCA2 T3085fs*26

CDKN2A rearrangement intron 2

MTOR A1459D

This assay tested >300 cancer-related genes, including the following 2 gene(s) routinely assessed in this tumor type: **KIT**, **PDGFRA**.

Report Highlights

- Variants with **diagnostic implications** that may indicate a specific cancer type: **KIT** exon 11 deletion (Q556_V559>H), exon 13 (V654A), exon 18 (A829P) (p. 6)
- Targeted therapies with **NCCN categories of evidence** for genomic findings in this patient's tumor type: **Regorafenib** (p. 9), **Nilotinib** (p. 13), **Ponatinib** (p. 14), **Sorafenib** (p. 14)
- Targeted therapies with **potential resistance** based on this patient's genomic findings: **Avapritinib** (p. 11), **Imatinib** (p. 11), **Sunitinib** (p. 12)
- High ctDNA Tumor Fraction was detected, indicating a lower risk of false negative results (p. 5)
- Clinical trial options based on this patient's genomic findings: (p. 15)

BIOMARKER FINDINGS

Blood Tumor Mutational Burden
3 Muts/Mb

ctDNA Tumor Fraction High (32%)

Microsatellite status

MSI-High Not Detected

GENOMIC FINDINGS

	VAF%
KIT exon 11 deletion (Q556_V559>H)	17.0%
exon 13 (V654A)	8.8%
exon 18 (A829P)	4.0%

10 Trials see p. 20

THERAPY AND CLINICAL TRIAL IMPLICATIONS

No therapies or clinical trials. See Biomarker Findings section

High ctDNA Tumor Fraction defined as $\geq 1.0\%$ based on concordance for defined short variants and fusions. See Biomarker Finding Summary.

MSI-High not detected. No evidence of microsatellite instability in this sample (see Appendix section).

THERAPIES WITH CLINICAL RELEVANCE (IN PATIENT'S TUMOR TYPE)	THERAPIES WITH CLINICAL RELEVANCE (IN OTHER TUMOR TYPE)
Regorafenib 1	Nilotinib 2A
Ripretinib ?	Ponatinib 2A
Avapritinib x	Sorafenib 2A
Imatinib x	Dasatinib
Sunitinib x	

x Extensive evidence showing variant(s) in this sample may confer resistance to this therapy

? Limited evidence showing variant(s) in this sample may confer resistance to this therapy

□ NCCN category for genomic findings in patient's tumor type

GENOMIC FINDINGS			VAF%	THERAPIES WITH CLINICAL RELEVANCE (IN PATIENT'S TUMOR TYPE)	THERAPIES WITH CLINICAL RELEVANCE (IN OTHER TUMOR TYPE)
BRCA2	T3085fs*26		1.5%	None	None
10 Trials see p. 15					
CDKN2A	rearrangement intron 2		12.8%	None	None
6 Trials see p. 18					
MTOR	A1459D		0.21%	None	None
4 Trials see p. 23					

 Extensive evidence showing variant(s) in this sample may confer resistance to this therapy

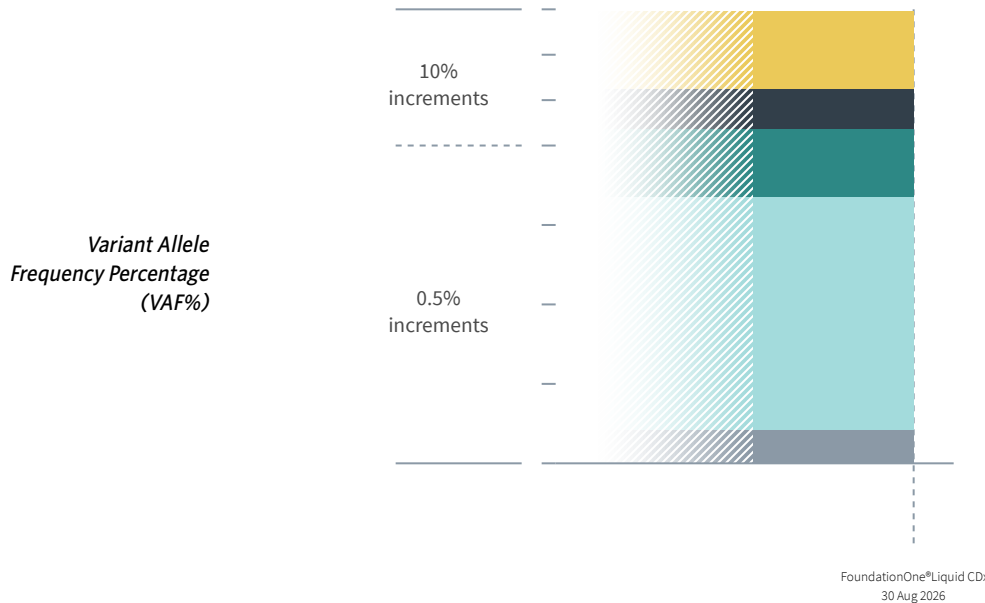
 Limited evidence showing variant(s) in this sample may confer resistance to this therapy

☐ NCCN category for genomic findings in patient's tumor type

NOTE Genomic alterations detected may be associated with activity of certain approved therapies; however, the therapies listed in this report may have varied clinical evidence in the patient's tumor type. Therapies and the clinical trials listed in this report may not be complete and/or exhaustive. Neither the therapies nor the trials identified are ranked in order of potential or predicted efficacy for this patient, nor are they ranked in order of level of evidence for this patient's tumor type. This report should be regarded and used as a supplementary source of information and not as the single basis for the making of a therapy decision. All treatment decisions remain the full and final responsibility of the treating physician and physicians should refer to approved prescribing information for all therapies. Therapies contained in this report may have been approved by the US FDA or other national authorities; however, they might not have been approved in your respective country. In the appropriate clinical context, germline testing of *APC*, *ATM*, *BAP1*, *BARD1*, *BRCA1*, *BRCA2*, *BRIP1*, *CDKN2A*, *CHEK2*, *FH*, *FLCN*, *MEN1*, *MLH1*, *MSH2*, *MSH6*, *MUTYH*, *NF1*, *NF2*, *PALB2*, *PMS2*, *POLD1*, *POLE*, *PTCH1*, *PTEN*, *RAD51C*, *RAD51D*, *RB1*, *RET*, *SDHA*, *SDHB*, *SDHC*, *SDHD*, *SMAD4*, *SMARCA4*, *SMARCB1*, *STK11*, *SUFU*, *TP53*, *TSC1*, *TSC2*, *VHL*, and *WT1* is recommended.

Variant Allele Frequency (VAF) is not applicable for copy number alterations, which include reportable focal amplifications and losses (homozygous deletions).

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HISTORIC PATIENT FINDINGS (Biomarker Findings)		ORD-2419716-01
Blood Tumor Mutational Burden		3 Muts/Mb
Microsatellite status		MSI-High Not Detected
ctDNA Tumor Fraction		32%
HISTORIC PATIENT FINDINGS (Genomic Findings)		VAF%
KIT	● exon 13 (V654A)	8.8%
	● exon 18 (A829P)	4.0%
	● exon 11 deletion (Q556_V559>H)	17.0%
BRCA2	● T3085fs*26	1.5%
CDKN2A	rearrangement intron 2	12.8%
MTOR	● A1459D	0.21%

IMPORTANT NOTE This comparison table refers only to genes and biomarkers assayed by prior FoundationOne®Liquid CDx or FoundationOne®CDx tests. Up to five previous tests may be shown.

For some genes in FoundationOne Liquid CDx, only select exons are assayed. Therefore, an alteration found by a previous test may not have been confirmed despite overlapping gene lists. Please refer to the Appendix for the complete list of genes and exons assayed. Variants reported for prior time points reflect reporting practices at the time of the historical test(s). Changes in variant reporting nomenclature, classification, or handling may result in the appearance of discrepancies across time points. The gene and biomarker list will be updated periodically to reflect new knowledge about cancer biology.

ctDNA Tumor Fraction may include previous Tumor Fraction results which reflect reporting practices at the time of reporting. Changes in biomarker reporting may result in the appearance of discrepancies across time points.

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As new scientific information becomes available, alterations that had previously been listed as Variants of Unknown Significance (VUS) may become reportable or reportable variants may become VUS.

Tissue Tumor Mutational Burden (TMB) and blood TMB (bTMB) are estimated from the number of synonymous and non-synonymous single-nucleotide variants (SNVs) and insertions and deletions (indels) per area of coding genome sampled, after the removal of known and likely oncogenic driver events and germline SNPs. Tissue TMB is calculated based on variants with an allele frequency of $\geq 5\%$, and bTMB is calculated based on variants with an allele frequency of $\geq 0.5\%$.

Not Tested = not baited, not reported on test, or test preceded addition of biomarker or gene

Not Detected = baited but not detected on test

Detected = present (VAF% is not applicable)

VAF% = variant allele frequency percentage

Cannot Be Determined = Sample is not of sufficient data quality to confidently determine biomarker status

Please note that other aspects of this table may have changed from the previous version to reflect the most up-to-date reporting information.

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BIOMARKER FINDINGS
BIOMARKER

Blood Tumor Mutational Burden

RESULT

3 Muts/Mb

POTENTIAL TREATMENT STRATEGIES
— Targeted Therapies —

On the basis of clinical evidence in solid tumors, increased blood TMB (bTMB) may be associated with greater sensitivity to immunotherapeutic agents, including anti-PD-L1¹⁻⁵, anti-PD-1^{2,5-8}, anti-PD-1/CTLA4 therapies^{2,7}, anti-PD-L1/CTLA4 therapies^{1,9-13}. A Phase 2 multi-solid-tumor trial showed that bTMB ≥ 16 Muts/Mb (as measured by this assay) was associated with improved survival from treatment with a PD-1 inhibitor alone or in combination with a CTLA-4 inhibitor⁷. In NSCLC, multiple clinical trials have shown patients with higher bTMB derive clinical benefit from immune

checkpoint inhibitors following single-agent or combination treatments with either CTLA4 inhibitors or chemotherapy, with reported high bTMB cutpoints ranging from 6 Muts/Mb-16 Muts/Mb^{3,8,11-13}. In HNSCC, a Phase 3 trial showed that bTMB ≥ 16 Muts/Mb (approximate equivalency ≥ 8 Muts/Mb as measured by this assay) was associated with improved survival from treatment with a PD-L1 inhibitor alone or in combination with a CTLA-4 inhibitor¹. In CRC, a Phase 2 study showed that bTMB ≥ 28 Muts/Mb (approximate equivalency ≥ 14 Muts/Mb as measured by this assay) was associated with improved OS from a PD-L1 inhibitor alone or in combination with a CTLA-4 inhibitor⁹⁻¹⁰.

FREQUENCY & PROGNOSIS

Among gastrointestinal stromal tumor cases profiled by F1LCDx, 3.8% of cases had a blood tumor mutational burden (bTMB) of >10 Muts/Mb¹⁴. Published data investigating the prognostic implications of TMB in GIST are limited (PubMed, Sep 2025).

FINDING SUMMARY

Blood tumor mutational burden (bTMB, also known as mutation load) is a measure of the number of somatic protein-coding base substitution and insertion/deletion mutations from circulating tumor DNA in blood. TMB is affected by a variety of causes, including exposure to mutagens such as ultraviolet light in melanoma¹⁵⁻¹⁶ and cigarette smoke in lung cancer¹⁷⁻¹⁸, treatment with temozolomide-based chemotherapy in glioma¹⁹⁻²⁰, mutations in the proofreading domains of DNA polymerases encoded by the POLE and POLD1 genes²¹⁻²⁵, and microsatellite instability (MSI)^{21,24-25}. High bTMB levels were not detected in this sample. It is unclear whether the bTMB levels in this sample would be predicted to be associated with sensitivity to PD-1- or PD-L1-targeting immune checkpoint inhibitors, alone or in combination with other agents^{3-4,8,26-27}. Depending on the clinical context, TMB testing of an alternate sample or by another methodology could be considered.

BIOMARKER

ctDNA Tumor Fraction

RESULT

High (32%)

POTENTIAL TREATMENT STRATEGIES
— Targeted Therapies —

Specimens with high circulating-tumor DNA (ctDNA) tumor fraction have high ctDNA content and thus higher sensitivity for identifying genomic alterations²⁸⁻²⁹. Such specimens are at a lower risk of false-negative results²⁸⁻³⁰. A negative result does not rule out the presence of a mutation below the limits of detection of the assay. Patients for whom no companion diagnostic alterations are detected should be considered for confirmation with a tumor tissue test that is FDA approved or

appropriately validated in other countries, if available. Single observations or changes over time of circulating-tumor DNA (ctDNA) quantity are not part of any clinical decision-making guidelines but may be a useful indicator for future cancer management³¹⁻³⁸.

FREQUENCY & PROGNOSIS

In a large genomic study of 25 solid tumor types, 69% of liquid biopsy samples had ctDNA levels $>1\%$ as measured by an investigational composite tumor fraction algorithm with a median tumor fraction of 2.2% across tumor types²⁹. Median ctDNA levels were reported to be highest in small cell lung cancer, liver, colon, and bladder tumor types and lowest in glioma and appendiceal cancers²⁹. Higher ctDNA levels were reported for patients with metastatic (Stage 4) tumors compared with patients with localized disease (Stages 1 to 3)³⁹. Higher ctDNA levels have been reported to be associated with worse prognosis in a variety of advanced solid tumors⁴⁰, including NSCLC⁴¹,

CRC⁴¹⁻⁴², pancreatic cancer⁴³, Ewing sarcoma and osteosarcoma⁴⁴, prostate cancer^{32,41,45}, breast cancer^{41,46}, leiomyosarcoma⁴⁷, esophageal cancer⁴⁸, and gastrointestinal cancer⁴⁹.

FINDING SUMMARY

The ctDNA tumor fraction provides an estimate of the percentage of circulating tumor DNA (ctDNA) present in a cell-free DNA (cfDNA) sample. The ctDNA tumor fraction algorithm utilized for FoundationOne Liquid CDx integrates multiple distinct genomic features, including aneuploidy and the observed allele frequencies of somatic short variants and rearrangements. High ctDNA tumor fraction was detected in this sample. In a study of patients with advanced NSCLC, the positive predictive agreement and negative predictive value of liquid biopsy compared with tissue for the detection of targetable driver alterations was 96% and 96%, respectively, when ctDNA tumor fraction was high (greater or equal to 1%)²⁸.

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GENOMIC FINDINGS
GENE
KIT
ALTERATION

exon 11 deletion (Q556_V559>H), exon 13 (V654A), exon 18 (A829P)

HGVS VARIANT

NM_000222.2:c.1668_1676del (p.Q556_V559delinsH),
NM_000222.2:c.1961T>C (p.V654A),
NM_000222.2:c.2485G>C (p.A829P)

VARIANT CHROMOSOMAL POSITION

chr4:55593601-55593610, chr4:55594258, chr4:55602664

POTENTIAL TREATMENT STRATEGIES
— Targeted Therapies —

On the basis of clinical evidence, primarily in GIST, melanoma, AML, and systemic mastocytosis, KIT activating alterations are associated with sensitivity to TKIs including imatinib, sunitinib, sorafenib, dasatinib, nilotinib, pazopanib, regorafenib, ponatinib, midostaurin, apatinib, avapritinib, and ripretinib⁵⁰⁻⁶⁴. In a Phase 1/2 dose escalation study of the Type-1 KIT-targeting agent bezucastinib monotherapy or in combination with sunitinib for patients with heavily pretreated KIT-mutated GIST after progressing on imatinib, sunitinib, or regorafenib, treatment with the combination therapy resulted in a median PFS (mPFS) of 12.1 months and a median OS (mOS) of 18.1 months⁶⁵. For 11 patients with detectable KIT mutations at enrollment, the combination resulted in 1 PR, 8 SD, and 2 PD; the subset of patients with KIT exon 9 or exon 11 baseline mutations and KIT secondary mutations in the exon 17 or exon 18 A-loop alone experienced 1 PR and 3 SDs, whereas the patients with additional exon 13 or exon 14 mutations experienced 1 SD and 2 PDs⁶⁵. Clinical benefit was also observed with bezucastinib monotherapy (mPFS 5.8 months and mOS 11.1 months), including 3 SDs and 1 PD for patients with KIT exon 11 and secondary KIT A-loop mutations, and 1 PD for 1 patient with exon 11 and a secondary KIT D654A mutation in exon 13⁶⁵. Avapritinib selectively inhibits mutated KIT and PDGFRA⁵⁵ and has demonstrated clinical activity against KIT exon 17 mutations, including KIT D816 mutations^{50,53,66-67}. In early phase clinical trials, 40% (10/25) of patients with heavily pretreated KIT-mutated GIST experienced tumor reduction⁵³ and 83% (19/23) of patients with advanced systemic mastocytosis responded⁶⁶. Preclinical⁶⁸⁻⁷⁰ and

limited clinical⁷¹⁻⁷³ data indicate that KIT A-loop mutations are sensitive to sorafenib, although preclinical evidence of potential resistance has been reported specifically for the D816V mutation^{69,74-75}. Several A-loop alterations at residues D816, D820, N822, and A829 have exhibited ponatinib sensitivity in preclinical evaluations⁷⁶⁻⁸¹. The use of mTOR inhibitors as an alternative therapeutic strategy has demonstrated limited success in KIT-mutated, imatinib-resistant melanoma, with 1 PR and 3 SD observed for 4 patients treated with everolimus⁸²⁻⁸³. However, no responses were observed for 10 patients with mastocytosis following everolimus monotherapy, with 8/10 patients harboring the KIT D816V mutation⁸⁴. The role of KIT amplification as a biomarker for response to mTOR inhibitors has not been investigated (PubMed, Feb 2026).

— Potential Resistance —

Mutation of residues within the KIT activation-loop (A-loop) has been associated with preclinical and clinical resistance to imatinib^{77,85-99} and sunitinib^{77,85,87,92,95,100} in patients with GIST. KIT V654A mutation has been associated with resistance to imatinib for patients and in cell-based models^{77,87,91-93,99,101-103}. KIT V654A is associated with resistance to avapritinib^{55,104-106} and was shown to reduce sensitivity to ripretinib based on clinical and preclinical data^{56,107-109} and preclinical¹¹⁰ as well as regorafenib and ponatinib in preclinical studies; this mutation still may be inhibited at clinically achievable doses⁷⁶⁻⁷⁷. Clinical and preclinical data suggest that mutations located within KIT exons 13 and 14, when co-occurring with other activating KIT mutations, reduce sensitivity to ripretinib^{56,107-109,111-115}.

FREQUENCY & PROGNOSIS

KIT mutation has been reported for 54-72% of patients with gastrointestinal stromal tumor (GIST), and the majority of these mutations affect exon 9 or exon 11¹¹⁶⁻¹¹⁸. It is estimated that up to 95% of GISTs overexpress the KIT protein, and KIT mutation has been reported to be an early event in the development of GIST^{116,119}. Mutations in KIT exon 11 are associated with a significantly greater response rate to imatinib, compared with individuals with mutations in exon 9 or without mutations in KIT¹²⁰⁻¹²¹. Studies comparing patients with GIST treated with imatinib with those treated with placebo indicate that the presence of exon 11 in-frame deletions is associated with a significantly

longer recurrence-free survival in univariate analysis; similar results were not observed for patients with exon 11 insertions, exon 11 point mutations, exon 9 mutations, or no KIT mutation¹²². Overall, patients with exon 11 mutations experienced a benefit from adjuvant imatinib therapy, which is in agreement with several previous studies¹²⁰⁻¹²³; although the reported effect on OS is favorable, it does not always reach statistical significance¹²²⁻¹²³. However, patients who develop imatinib resistance have a poorer prognosis than those who do not, including a reduced 5-year survival rate¹²⁴.

FINDING SUMMARY

KIT (also called c-KIT) encodes a receptor tyrosine kinase that signals through the PI3K-AKT and RAS-MAPK pathways¹²⁵⁻¹²⁶. KIT alterations in exon 11 leading to the disruption of the autoinhibitory domain (amino acids 545-579)¹²⁷⁻¹³⁴ or ITD proximal to the kinase domain (amino acids 569-591)¹³⁵, such as observed here, are predicted to be activating and sensitive to imatinib^{120,136-137}. KIT exon 11 deletions are frequently observed in the context of gastrointestinal stromal tumor (GIST)^{86,121,129} and KIT-ITDs in acute myeloid leukemia (AML)^{136,138-141}. Extensive clinical and preclinical studies have reported that KIT mutations in exon 17 within the activation loop (A-loop), including C809, D816, D820, N822, Y823, and A829, have emerged as secondary alterations conferring resistance to imatinib^{77,86-97}, or have shown resistance to sunitinib^{77,87,92,142}. However, KIT A-loop mutations have shown clinical sensitivity to avapritinib^{50,53,55,66-67}. The V654A exon 13 mutation is located within the kinase domain of KIT and is considered activating¹⁴³. It is also a common secondary mutation, acquired in addition to an existing activating mutation in KIT; this mutation has been associated with resistance to imatinib^{77,87,91-93,101-102} and avapritinib^{55,104-105} clinically. Based on clinical (2023; AACR Abstract 3887)^{56,107-109} data, V654A mutations may be associated with reduced sensitivity to ripretinib; preclinical models indicate decreased sensitivity to regorafenib and ponatinib⁷⁶⁻⁷⁷.

POTENTIAL DIAGNOSTIC IMPLICATIONS

KIT mutations are characteristic drivers of GIST and may aid in guiding targeted therapy approaches in this setting (NCCN GIST Guidelines, v1.2025)¹⁴⁴.

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GENOMIC FINDINGS
GENE
BRCA2
ALTERATION

T3085fs*26

HGVS VARIANT

NM_000059.3:c.9253dup (p.T3085Nfs*26)

VARIANT CHROMOSOMAL POSITION

chr13:32954272

POTENTIAL TREATMENT STRATEGIES
— Targeted Therapies —

Alterations that inactivate BRCA1 or BRCA2 may confer sensitivity to PARP inhibitors¹⁴⁵⁻¹⁶³ or ATR inhibitors¹⁶⁴⁻¹⁶⁹. Clinical responses to PARP inhibitors have been reported for patients with either germline or somatic BRCA1/2 mutations mostly in ovarian^{152,155,162,170}, breast^{149,171-172}, prostate^{147,173-174}, and pancreatic cancer¹⁶³, including patients with platinum-resistant or -refractory disease^{146,151,158,161}. Among other tumor types, although 1 complete response to olaparib has been reported for 1/3 patients with BRCA2-mutated colon adenocarcinoma from a basket trial¹⁷⁵, Phase 2 and retrospective studies of PARP inhibitors did not find clinical benefit for 14 patients with BRCA1- or BRCA2-mutated CRC¹⁷⁵⁻¹⁷⁸. In the Phase 2 KEYLYNK-007 trial, patients with HRRm/HRD advanced tumors were treated with the combination olaparib and pembrolizumab, and the cohort of patients with BRCA mutated cancer, outside of ovarian and breast cancer, reported improved ORR of 27% (36/132) vs. 12% (12/104), mPFS of 4.4 vs. 3.7 months, and mOS of 14.0 vs.

10.4 months compared with patients with non-BRCA mutated disease¹⁷⁹. In a Phase 2 study for patients with PARP-resistant ovarian cancer, the combination of the WEE1 inhibitor adavosertib and olaparib elicited improved clinical benefit (ORR 29%; DCR 89%) than adavosertib alone (ORR 23%; DCR 63%); however, among patients with BRCA-mutated tumors, clinical benefit was comparable between the combination (ORR 19%) and monotherapy (ORR 20%)¹⁸⁰. A Phase 1 monotherapy trial of adavosertib for BRCA1/2-mutated solid tumors reported PRs in 2 patients (ovarian serous and oral SCC), while a third patient with BCRA1- and TP53-mutated ovarian serous carcinoma experienced 14% tumor shrinkage prior to disease progression¹⁸¹. Limited clinical and preclinical evidence support the sensitivity of BRCA2 inactivating mutations to ATR inhibitors, including two responses to berzosertib for patients with BRCA2 inactivating alterations in neuroendocrine prostate cancer and breast cancer^{169,182-185}.

— Nontargeted Approaches —

Inactivation of BRCA2 may also predict sensitivity to DNA-damaging drugs such as trabectedin, lurbinectedin, and the platinum chemotherapies cisplatin and carboplatin¹⁸⁶⁻¹⁹⁸.

FREQUENCY & PROGNOSIS

In a genomic study of sarcoma, BRCA2 mutation or homozygous deletion was not observed in any of 22 gastrointestinal stromal tumors (GIST)¹⁹⁹. A case report has described a patient with a BRCA2 mutation who developed GIST, as well as other tumors²⁰⁰. Published data investigating the

prognostic implications of BRCA2 alterations in GIST are limited (PubMed, Jan 2025).

FINDING SUMMARY

The BRCA2 tumor suppressor gene encodes a protein that regulates the response to DNA damage²⁰¹. Inactivating BRCA2 alterations can lead to homologous recombination deficiency²⁰². Alterations such as seen here may disrupt BRCA2 function or expression^{201,203-218}.

POTENTIAL GERMLINE IMPLICATIONS

One or more of the BRCA2 variants observed here has been described in the ClinVar database as a likely pathogenic or pathogenic germline mutation (by an expert panel or multiple submitters)²¹⁹. Follow-up germline testing would be needed to distinguish whether the finding in this patient is somatic or germline. Inactivating germline mutations in BRCA1 or BRCA2 are associated with autosomal dominant hereditary breast and ovarian cancer²²⁰⁻²²¹, and the lifetime risk of breast and ovarian cancer in BRCA2 mutation carriers has been estimated to be as high as >80% and 23%, respectively²²². Elevated risk for other cancer types, including gastric, pancreatic, prostate, and colorectal, has also been identified, with an increase in risk ranging from 20 to 60%²²³. The estimated prevalence of deleterious germline BRCA1/2 mutations in the general population is between 1:400 and 1:800, with an approximately 10-fold higher prevalence in the Ashkenazi Jewish population^{222,224-229}. In the appropriate clinical context, germline testing of BRCA2 is recommended.

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GENOMIC FINDINGS
GENE
CDKN2A
ALTERATION

rearrangement intron 2

REARRANGEMENT DETAILS

CDKN2A(NM_000077) rearrangement intron 2

POTENTIAL TREATMENT STRATEGIES
— Targeted Therapies —

Preclinical and clinical data suggest that tumors with CDKN2A alterations may predict sensitivity to CDK4/6 inhibitors, such as abemaciclib, ribociclib, and palbociclib²³⁰⁻²⁴⁵. Clinical benefit from CDK4/6 inhibitors has been observed for patients with B-cell lymphoma, head and neck cancer, thymic cancer, NSCLC, sarcoma, mesothelioma, and several other tumor types^{230-232,237-247}. Although preclinical studies have suggested that loss of p14ARF function may be associated with reduced sensitivity to MDM2 inhibitors²⁴⁸⁻²⁴⁹, the clinical relevance of p14ARF

as a predictive biomarker is not clear.

FREQUENCY & PROGNOSIS

Putative concurrent homozygous deletion of CDKN2A and CDKN2B was seen in 14% (3/22) of gastrointestinal stromal tumors (GISTs) in the Sarcoma Genome Project dataset¹⁹⁹. Loss of at least part of chromosomal region 9p21, which contains the CDKN2A/B loci, was reported in 63% of patient GISTs²⁵⁰. Numerous studies have shown that loss of expression of p16INK4a and p14ARF correlates with aggressive disease, worse prognosis, and shorter survival in patients with GIST²⁵⁰⁻²⁵⁵.

FINDING SUMMARY

CDKN2A encodes two different, unrelated tumor suppressor proteins, p16INK4a and p14ARF²⁵⁶⁻²⁵⁷. Inactivation of p16INK4a leads to dysregulation of the CDK4/6-cyclin-Rb pathway and loss of cell cycle control²⁵⁸⁻²⁵⁹. The tumor suppressive functions of p14ARF involve stabilization and activation of p53, via a mechanism of MDM2 inhibition²⁶⁰⁻²⁶¹. One or more alterations observed here are predicted to result in p16INK4a loss of

function²⁶²⁻²⁸³. One or more alterations seen here are predicted to result in p14ARF loss of function^{266,283-286}.

POTENTIAL GERMLINE IMPLICATIONS

Germline CDKN2A mutation is associated with melanoma-pancreatic cancer syndrome, a condition marked by increased risk of developing malignant melanoma and/or pancreatic cancer²⁸⁷. Mutation carriers within families may develop either or both types of cancer, and melanoma cases may be referred to as familial or hereditary melanoma²⁸⁸⁻²⁸⁹. CDKN2A is the most implicated gene in familial melanoma, with germline mutations present in 16% to 20% of familial melanoma cases²⁹⁰⁻²⁹². CDKN2A alteration has also been implicated in familial melanoma-astrocytoma syndrome, an extremely rare tumor association characterized by dual predisposition to melanoma and nervous system tumors²⁹³⁻²⁹⁵. In the appropriate clinical context, germline testing of CDKN2A is recommended.

GENE
MTOR
ALTERATION

A1459D

HGVS VARIANT

NM_004958.3:c.4376C>A (p.A1459D)

VARIANT CHROMOSOMAL POSITION

chr1:11217302

been reported in a subset of patients with MTOR-mutated solid tumors, most notably RCC²⁹⁸⁻³⁰¹, as well as individual cases of gastric cancer and angiosarcoma²⁹⁸. However, multiple retrospective and prospective studies, including NCI-MATCH and analyses of the RECORD-3, GRANITE-1, and EVOLVE-1 trials, have not demonstrated a consistent association between MTOR pathway alterations and clinical benefit from everolimus³⁰²⁻³⁰⁴.

FREQUENCY & PROGNOSIS

MTOR mutations have been reported in 1.2% of gastrointestinal stromal tumors (GIST)³⁰⁵. Published data investigating the prognostic implications of MTOR alterations in GIST are limited (PubMed, Feb 2026). However, the PI3K/

mTOR/AKT pathway is frequently activated in GIST tumors, and strategies for targeting this pathway are under investigation³⁰⁶⁻³¹⁰. Activation of the PI3K/mTOR/AKT pathway was reported to be more common in GIST with acquired resistance to imatinib (53%) when compared to imatinib-sensitive (27%) and primary resistant GIST (33%)³⁰⁶.

FINDING SUMMARY

mTOR encodes an intracellular serine/threonine kinase that integrates upstream signals, including those from the PI3K-AKT pathway to regulate cellular growth, metabolism, and survival³¹¹⁻³¹². MTOR mutations such as observed here have shown to be activating and may confer sensitivity to mTOR inhibitors^{296-297,313-320}.

POTENTIAL TREATMENT STRATEGIES
— Targeted Therapies —

Activating MTOR mutations activate PI3K-AKT-mTOR signaling²⁹⁶⁻²⁹⁷. Clinical responses to rapalogs such as everolimus and temsirolimus have

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THERAPIES WITH CLINICAL BENEFIT
IN PATIENT'S TUMOR TYPE

Regorafenib

Assay findings association

KIT

 exon 11 deletion (Q556_V559>H),
 exon 13 (V654A), exon 18
 (A829P)

AREAS OF THERAPEUTIC USE

Regorafenib is a small-molecule inhibitor of multiple kinases, including RET, VEGFRs, PDGFRs, KIT, and RAF family proteins. It is FDA approved to treat hepatocellular carcinoma that has been previously treated with sorafenib, metastatic colorectal cancer (CRC), or advanced gastrointestinal stromal tumors (GISTs). Please see the drug label for full prescribing information.

GENE ASSOCIATION

On the basis of clinical evidence in gastrointestinal stromal tumors³²¹⁻³²² and melanoma⁵⁴, activating KIT alterations may predict sensitivity to regorafenib. Limited preclinical data has shown that KIT V654A may reduce sensitivity to regorafenib⁷⁷. In a study of patients with GIST, V654A emerged during regorafenib treatment in 4 patients and increased in allele frequency in 4 additional patients³²³.

SUPPORTING DATA

In the Phase 3 VOYAGER trial for patients with KIT-mutated GIST in the third- or fourth-line setting, regorafenib was superior to avapritinib (median PFS

[mPFS]: 5.6 months vs. 4.2 months, HR=0.80)³²⁴.

Regorafenib was superior to avapritinib for patients with mutations in exons 13 or 14 (mPFS: 7.4 months vs. 1.9 months) or exons 17 or 18 (6.7 months vs. 4.7 months); mPFS for patients without secondary mutations in KIT exons 13 or 14 was similar between the 2 groups (5.6 months vs. 5.6 months), but avapritinib was inferior to regorafenib for patients with mutations in exons 13 or 14 (1.9 months vs. 7.4 months) or exons 17 or 18 (4.7 months vs. 6.7 months)³²⁵. A Phase 2 follow-up study of regorafenib treatment for patients with advanced GIST and a KIT exon 11 mutation reported a longer median PFS (13.4 months) compared with patients with KIT/PDGFR wild-type, non-SDH-deficient tumors (1.6 months)³²¹.

Regorafenib has shown efficacy in the third line for advanced gastrointestinal stromal tumors (GISTs) based on a Phase 3 randomized study comparing regorafenib to placebo; patients who had progressed on imatinib and sunitinib demonstrated significantly longer median PFS when treated with regorafenib (4.8 vs. 0.9 months)⁵⁹. A Phase 2 study of regorafenib in the second-line setting for patients with imatinib-resistant advanced GIST reported a median PFS of 36.3 weeks³²⁶.

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THERAPIES ASSOCIATED WITH UNCLEAR RESISTANCE

IN PATIENT'S TUMOR TYPE

Ripretinib

Resistance of variant(s) to associated therapy is unclear

Assay findings association

KIT

exon 11 deletion (Q556_V559>H),
exon 13 (V654A), exon 18
(A829P)

AREAS OF THERAPEUTIC USE

Ripretinib is a selective kinase inhibitor of PDGFRA and KIT. It is FDA approved to treat patients with gastrointestinal stromal tumor (GIST). Please see the drug label for full prescribing information.

GENE ASSOCIATION

On the basis of clinical evidence in KIT-mutated GIST⁵¹ as well as strong preclinical evidence¹⁰⁶, KIT mutation may predict sensitivity to ripretinib. On the basis of clinical and preclinical evidence, mutations within KIT exons 13 and 14, when co-occurring with other activating KIT mutations, reduce sensitivity to ripretinib^{56,107-109,111-115}. In the Phase 3 INTRIGUE trial of ripretinib versus sunitinib in the second line setting, patients with exon 11 co-mutated with exon 13 or exon 14 mutations experienced inferior mPFS on ripretinib (4.0 months vs. 15.0 months, HR=3.94); however, patients with triplet mutations in exon 11, exon 13 or exon 14, and exon 17 or exon 18 did not experience significantly poorer mPFS relative to patients with triplet mutations treated with sunitinib (8.1 months vs. 10.9 months, HR=1.07)¹⁰⁸. Investigators reported frequent emergence of KIT exon 13 and 14 mutations at progression on ripretinib treatment¹¹¹. Preclinical data suggest that KIT exon 13 and 14 alterations in the absence of other KIT activating mutations may still be sensitive to ripretinib treatment^{106,111,327}. Therefore, it is

unclear whether ripretinib would be beneficial in this context.

SUPPORTING DATA

The Phase 3 INVICTUS study comparing ripretinib with placebo for patients with previously treated advanced gastrointestinal stromal tumor (GIST) reported improved median PFS (mPFS; 6.3 vs. 1.0 months, HR=0.15), median OS (15.1 vs. 6.6 months, HR=0.36), and ORR (9.4% vs. 0%)³²⁸. A Phase 1 study evaluating ripretinib in previously treated GIST also reported ORRs of 24%, 24%, or 9% as second-line, third-line, or fourth-or-later-line therapies, respectively⁵¹. Inpatient dose escalation of ripretinib after initial relapse provided a further mPFS of 3.3-5.6 months in these studies without additional toxicity³²⁹⁻³³⁰. The Phase 3 INTRIGUE trial comparing ripretinib with sunitinib in advanced GIST after relapse on imatinib reported similar mPFS (8.0 vs. 8.3 months), but ripretinib was associated with a more favorable safety profile as well as a higher ORR for patients with KIT exon 11 mutations (24% for ripretinib compared with 15% for sunitinib)³³¹. Outcomes for patients treated with ripretinib in the INTRIGUE trial differed by co-occurring KIT mutation status: exon 11 and exon 13 or 14 mutations were associated with shorter PFS (4 months) than those with exon 11 and exon 17 or 18 mutations (14.2 months)¹⁰⁸.

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THERAPIES ASSOCIATED WITH RESISTANCE

IN PATIENT'S TUMOR TYPE

Avapritinib

✖ Resistance of variant(s) to associated therapy is likely

Assay findings association

KIT

exon 11 deletion (Q556_V559>H),
exon 13 (V654A), exon 18
(A829P)

AREAS OF THERAPEUTIC USE

Avapritinib is a selective kinase inhibitor of PDGFRA and KIT activating mutations. It is FDA approved to treat unresectable or metastatic gastrointestinal stromal tumor (GIST) that is positive for PDGFRA mutation in exon 18, including the resistance mutation D842V. It is also approved to treat indolent or advanced systemic mastocytosis. Please see the drug label for full prescribing information.

GENE ASSOCIATION

On the basis of prospective Phase 1 and Phase 2 studies in KIT-mutated GIST^{53,55} and systemic mastocytosis (SM)³³²⁻³³³, as well as a case report in metastatic vulvar melanoma³³⁴, KIT mutation may predict sensitivity to avapritinib. Avapritinib has shown clinical activity against the KIT A-loop mutation D816V in a Phase 1 trial for SM³³⁵, and against additional KIT A-loop mutations in preclinical studies⁵⁵. Based on a clinical study for patients with gastrointestinal stromal tumor (GIST)¹⁰⁴ and preclinical data^{55,106}, KIT V654A, when co-occurring with KIT activating alteration, is associated with resistance to avapritinib. The relevance of this therapeutic strategy for cases in which V654A alteration is seen alone is unclear.

SUPPORTING DATA

In the Phase 3 VOYAGER trial for patients with KIT-mutated GIST in the third- or fourth-line setting, avapritinib was not superior to regorafenib (median PFS [mPFS] of 4.2 vs. 5.6 months, HR=1.25)³²⁴; mPFS for patients without secondary mutations in the KIT ATP binding pocket (ABP, exons 13 or 14) was similar between the 2 groups (5.6 vs. 5.6 months), but avapritinib was inferior to regorafenib for patients with ABP mutations (1.9 vs. 7.4 months) or activation loop (AL, exons 17 or 18; 4.7 vs. 6.7 months)³²⁵. Post hoc analysis of Phase 1 and Phase 1/2 avapritinib trials showed improved outcomes for patients with GIST both positive for KIT AL mutations and negative for KIT ABP mutations relative to patients with GIST and other KIT mutations (ORR of 27% vs. 12%; mPFS of 9.1 vs. 3.5 months)³³⁶. Avapritinib has primarily been studied in the context of PDGFRA- and KIT-mutated gastrointestinal stromal tumor (GIST)^{324,336-339} and KIT-mutated mastocytosis^{50,66,332-333,335}. In the Phase 1 NAVIGATOR trial in heavily pretreated GIST, patients with KIT (n=110) or PDGFRA non-D842V (n=3) alterations experienced a median PFS of 3.7 months and median OS of 11.6 months³³⁷; patients with PDGFRA D842V-positive GIST experienced an ORR of 88% (49/56; 5 CRs, 44 PRs), a 12-month PFS rate of 81%, and a 24-month OS rate of 81%³³⁸.

Imatinib

✖ Resistance of variant(s) to associated therapy is likely

Assay findings association

KIT

exon 11 deletion (Q556_V559>H),
exon 13 (V654A), exon 18
(A829P)

AREAS OF THERAPEUTIC USE

Imatinib targets the BCR-ABL fusion protein, PDGFR, and KIT. It is FDA approved to treat KIT-positive gastrointestinal stromal tumors (GISTs), Philadelphia chromosome-positive chronic myeloid leukemia and acute lymphoblastic leukemia, myelodysplastic syndrome/myeloproliferative syndrome, aggressive systemic mastocytosis without a D816V KIT mutation, hypereosinophilic syndrome and/or chronic eosinophilic leukemia, and dermatofibrosarcoma protuberans. Please see the drug label for full prescribing information.

GENE ASSOCIATION

On the basis of clinical and preclinical data in KIT-mutated^{137,340-342}, KIT-amplified^{340,342-344}, or KIT-expressing tumors^{141,345-351}, KIT activating alterations may confer sensitivity to imatinib. Mutation of residues within or adjacent to the KIT activation loop, as observed here, have demonstrated resistance to imatinib in patients with GIST^{86,91,93,95,98-99,102-103,352-354}. The KIT V654A mutation is predicted to confer resistance to imatinib based on strong

clinical evidence in GIST^{86,91,93,95,98-99,103,352-353,355}.

SUPPORTING DATA

Clinical studies in GIST have reported that imatinib treatment results in significantly longer recurrence-free survival (RFS) and OS^{123,345}. The Phase 3 BFR14 trial reported improved clinical outcomes for patients with GIST who continued treatment with imatinib after achieving disease control compared with patients who discontinued treatment at disease control³⁵⁶. A meta-analysis of imatinib clinical trials suggests that patients with GIST harboring KIT exon 11 mutations respond better to imatinib than those without³⁵⁷. Several Phase 3 trials of imatinib, including a 10-year outcomes study, reported significantly increased OS in patients with GIST harboring KIT exon 11 mutations relative to those with KIT exon 9 mutations³⁵⁸⁻³⁵⁹. A case study has reported a PR and 5-year disease stabilization upon treatment with imatinib in a patient with GIST harboring a PDGFRA exon 12 mutation³⁶⁰.

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THERAPIES ASSOCIATED WITH RESISTANCE

IN PATIENT'S TUMOR TYPE

Sunitinib

✖ Resistance of variant(s) to associated therapy is likely

Assay findings association

KIT

exon 11 deletion (Q556_V559>H),
exon 13 (V654A), exon 18
(A829P)

AREAS OF THERAPEUTIC USE

Sunitinib is a small-molecule TKI that targets PDGFRs, VEGFRs, KIT, FLT3, CSF-1R, and RET. It is FDA approved to treat advanced or metastatic pancreatic neuroendocrine tumors, gastrointestinal stromal tumors (GISTs) for patients who have progressed on or are intolerant to imatinib, and advanced renal cell carcinoma (RCC) as well as for the adjuvant treatment of patients at high risk of recurrent RCC after nephrectomy. Please see the drug label for full prescribing information.

GENE ASSOCIATION

On the basis of clinical and preclinical data in KIT-mutated^{92,361-365} or KIT-expressing tumors^{361,366}, KIT activating alterations may predict sensitivity to sunitinib. On the basis of retrospective analyses^{92,100}, case studies³⁶⁷, and preclinical evidence in GIST^{77,92,367}, KIT V654A is predicted to be sensitive to sunitinib. Mutation of residues within the KIT activation loop (A-loop) is predicted to confer resistance to sunitinib based on extensive clinical data in gastrointestinal stromal tumor^{87,92,95,100}.

SUPPORTING DATA

In a Phase 1/2 trial of sunitinib, patients with imatinib-resistant GIST harboring KIT exon 11 mutations experienced worse median OS (mOS, 12.3 months) and clinical benefit rate (34%) than patients with exon 9 mutations (26.9 months; 58%) or wildtype KIT (30.5

months; 56%)⁹²; secondary mutations in KIT V654 in exon 13 or T670 in exon 14 were associated with numerically improved mOS compared with patients carrying secondary activation-loop mutations in exon 17 or exon 18 (13.0 months vs. 4.0 months, n=18 and n=13)⁹². The presence of secondary mutations may have impaired the response of patients with secondary mutations; KIT V560D in exon 11 was potentially inhibited by sunitinib in vitro in the absence of secondary KIT mutations⁹². Biomarker analysis from a Phase 1/2 study of sunitinib in combination with investigational KIT inhibitor bezuglactinib reported benefit in later lines for patients with similar exon 17 or 18 secondary alterations⁶⁵. In a retrospective analysis of second-line sunitinib, patients with advanced gastrointestinal stromal tumor (GIST) receiving sunitinib experienced superior median OS (mOS) compared with patients receiving any other second-line treatment (32.4 vs. 27.1 months, n=436 and n=74)³⁶⁸. The median time to progression for patients with imatinib-refractory or -intolerant GIST treated with sunitinib was 27.3 weeks, as compared with 6.4 weeks for patients treated with placebo³⁶⁹. A Phase 1/2 trial of sunitinib in combination with the investigational KIT-inhibitor bezuglactinib for a population heavily pretreated with KIT-targeted agents reported a median PFS and mOS of 12.1 months and 18.1 months, respectively; the study observed 1 CR, 2 PRs, and a clinical benefit rate of 80% (n=15)⁶⁵.

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THERAPIES WITH CLINICAL BENEFIT

IN OTHER TUMOR TYPE

Dasatinib

Assay findings association

KIT

exon 11 deletion (Q556_V559>H),
exon 13 (V654A), exon 18
(A829P)

AREAS OF THERAPEUTIC USE

Dasatinib is a kinase inhibitor that targets the BCR-ABL fusion protein, SRC family kinase receptors, KIT, EPHA2, and PDGFR-beta. It is FDA approved to treat certain subtypes of Philadelphia chromosome-positive (Ph+) chronic myeloid leukemia and Ph+ acute lymphoblastic leukemia. Please see the drug label for full prescribing information.

GENE ASSOCIATION

On the basis of clinical evidence in melanoma³⁷⁰⁻³⁷¹ and preclinical evidence in other cancer types³⁷²⁻³⁷³, activating KIT alterations may confer sensitivity to dasatinib. Preclinical and limited clinical data indicate that secondary imatinib-resistant mutations, including V654A and A-loop alterations at residues D816, D820, N822, and

A829, remain sensitive to dasatinib^{68,373-377}.

SUPPORTING DATA

A Phase 2 trial of dasatinib for patients with TKI-naïve gastrointestinal stromal tumors (GIST) reported an ORR of 74% (31/42, 14 CR) and a median PFS (mPFS) of 13.6 months³⁷⁸. A Phase 2 trial of dasatinib for patients with GIST who were refractory to imatinib and sunitinib reported an mPFS of 3.1 months, and 3.4% (2/58) experienced PRs³⁷⁹; a similar Phase 2 trial reported a PR rate of 32% (15/47), with 21% of patients demonstrating PFS for >6 months by Choi criteria³⁸⁰. A trial of dasatinib for patients with GIST previously treated with imatinib reported a 25% ORR (12/48, all PR), mPFS of 2.9 months, and OS of 19 months by Choi criteria³⁸¹.

Nilotinib

Assay findings association

KIT

exon 11 deletion (Q556_V559>H),
exon 13 (V654A), exon 18
(A829P)

AREAS OF THERAPEUTIC USE

Nilotinib targets tyrosine kinases such as ABL (including BCR-ABL), PDGFRs, KIT, CSF1R, DDR1, and DDR2. It is FDA approved to treat newly diagnosed pediatric or adult patients with Philadelphia chromosome-positive (Ph+) chronic myeloid leukemia (CML) in chronic phase, adults with Ph+ CML in chronic or accelerated phase with resistance or intolerance to prior therapy including imatinib, and pediatric patients with Ph+ CML in chronic phase with resistance or intolerance to prior tyrosine-kinase inhibitor therapy. Please see the drug label for full prescribing information.

GENE ASSOCIATION

Additionally, imatinib-resistant KIT mutations such as V654A and A-loop alterations at residues D816, D820, N822, and A829 maintain sensitivity to nilotinib as demonstrated in multiple clinical and preclinical studies^{68,143,374-375,382-384}. On the basis of clinical and preclinical data in KIT-mutated³⁸⁴⁻³⁸⁸, KIT-amplified³⁸⁵, or

KIT-expressing tumors³⁸⁹, KIT activating alterations may confer sensitivity to nilotinib.

SUPPORTING DATA

The Phase 3 ENESTg3 trial of nilotinib for 240 patients with advanced gastrointestinal stromal tumors (GIST) whose cancer had progressed on prior imatinib and/or sunitinib treatment showed no significant difference in PFS between nilotinib and best supportive care, but patients treated with nilotinib showed increased OS³⁹⁰. The Phase 3 ENESTg1 trial of nilotinib versus imatinib as a first-line treatment for patients with advanced GIST did not show clinical improvement for nilotinib; longer OS and PFS were reported for imatinib in the full population of patients (397), but patients with KIT exon 11 mutations saw similar PFS in both treatment groups³⁸⁷. A Phase 2 clinical trial showed that nilotinib was well tolerated and suggested that it may be particularly useful in treating a subset of GIST with KIT exon 17 mutations³⁹¹.

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THERAPIES WITH CLINICAL BENEFIT

IN OTHER TUMOR TYPE

Ponatinib

Assay findings association

KIT

exon 11 deletion (Q556_V559>H),
exon 13 (V654A), exon 18
(A829P)

AREAS OF THERAPEUTIC USE

Ponatinib is a multikinase inhibitor targeting BCR-ABL, RET, KIT, FLT-3, PDGFRs, VEGFRs, FGFRs, and other tyrosine kinases. It is FDA approved in combination with chemotherapy to treat Philadelphia chromosome-positive (Ph+) acute lymphoblastic leukemia (ALL) or as a monotherapy to treat advanced, T315I-mutated chronic myeloid leukemia (CML) and Ph+ ALL, as well as for CML and Ph+ ALL patients for whom no other TKI is indicated. Please see the drug label for full prescribing information.

GENE ASSOCIATION

KIT exon 9 and 11 activating mutations have demonstrated sensitivity to ponatinib in several preclinical cell models^{76,78-81,392}, with clinical efficacy demonstrated in patients with GIST exhibiting exon 11 mutations^{77,393} and imatinib resistance⁵². KIT V654A has been shown to result in decreased sensitivity to ponatinib in a preclinical study⁷⁶⁻⁷⁷. Additionally, in a study of

patients with GIST, KIT V654A emerged during treatment in 8 patients³⁹⁴.

SUPPORTING DATA

Data from a Phase 2 trial in gastrointestinal stromal tumors (GISTs) indicated that ponatinib induced a clinical benefit rate (CBR; patients exhibiting CR, PR, or SD for at least 16 weeks) in 36% (10/28) of patients with KIT exon 11 mutations compared with a CBR of 20% (3/15) for patients lacking mutations in exon 11³⁹⁴. Ponatinib has also demonstrated robust preclinical activity in patient-derived GIST cell lines against multiple imatinib-resistant KIT mutations, including the gatekeeper residue, T670I, and secondary mutations within the activation loop, including those at residues D816, D820, N822, and A829⁷⁷. Moreover, ponatinib displayed clinical activity in 2/3 patients with refractory GIST, all of whom had been previously treated at a minimum with imatinib, sunitinib, and regorafenib⁷⁷.

Sorafenib

Assay findings association

KIT

exon 11 deletion (Q556_V559>H),
exon 13 (V654A), exon 18
(A829P)

AREAS OF THERAPEUTIC USE

Sorafenib is a kinase inhibitor that targets the RAF kinases, KIT, FLT3, RET, VEGFRs, and PDGFRs. It is FDA approved to treat unresectable hepatocellular carcinoma, advanced renal cell carcinoma (RCC), and recurrent or metastatic differentiated thyroid carcinoma. Please see the drug label for full prescribing information.

GENE ASSOCIATION

On the basis of clinical and preclinical data in KIT-mutated^{68,70-72,395-398} or KIT-expressing tumors³⁹⁹⁻⁴⁰², KIT activating alterations may predict sensitivity to sorafenib. On the basis of preclinical⁶⁸⁻⁷⁰ and limited clinical⁷¹⁻⁷³ data, mutations of residues within the KIT activation loop (A-loop) are predicted to be sensitive to sorafenib, although preclinical evidence of potential resistance has been reported specifically for the D816V mutation^{69,74-75}.

SUPPORTING DATA

Phase 2 studies of sorafenib in patients with gastrointestinal stromal tumors (GIST) previously treated with imatinib and sunitinib have reported PRs in 10-13% of patients and SDs in 52-57% of patients^{397,403}. A retrospective analysis of sorafenib in 124 patients with GIST previously treated with imatinib and sunitinib reported that 57% (70/124) of patients achieved disease stabilization and 10% (12/124) achieved PR⁴⁰⁴. A real-world evidence study of patients with advanced GIST treated with sorafenib reported median overall survival (mOS) of 17.4 months⁴⁰⁵. One study of 60 patients who had progressed on imatinib or sunitinib reported 3 PRs to sorafenib treatment, and median PFS (mPFS) of 7.7 months and mOS of 13.5 months⁴⁰⁶.

NOTE Genomic alterations detected may be associated with activity of certain US FDA or other specific country approved therapies; however, the therapies listed in this report may have varied evidence in the patient's tumor type. The listed therapies are not ranked in order of potential or predicted efficacy for this patient or in order of level of evidence for this patient's tumor type. The therapies listed in this report may not be complete and/or exhaustive. Furthermore, the listed therapies are limited to US FDA approved pharmaceutical drug products that are linked to a specific genomic alteration. There may also be US FDA approved pharmaceutical drug products that are not linked to a genomic alteration. Further there may also exist pharmaceutical drug products that are not approved by the US FDA or other national authorities. There may also be other treatment modalities available than pharmaceutical drug products.

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CLINICAL TRIALS

IMPORTANT Clinical trials are ordered by gene and prioritized by: age range inclusion criteria for pediatric patients, proximity to ordering medical facility, later trial phase, and verification of trial information within the last two months. While every effort is made to ensure the accuracy of the information contained below, the information available in the public domain is continually updated and should be investigated by the physician or

research staff. This is not a comprehensive list of all available clinical trials. There may also be compassionate use or early access programs available, which are not listed in this report. Foundation Medicine displays a subset of trial options and ranks them in this order of descending priority: Qualification for pediatric trial → Geographical proximity → Later trial phase. Clinical trials are not ranked in order of potential or predicted efficacy for this patient or

in order of level of evidence for this patient's tumor type. Clinical trials listed here may have additional enrollment criteria that may require medical screening to determine final eligibility. For additional information about listed clinical trials or to conduct a search for additional trials, please see clinicaltrials.gov. However, clinicaltrials.gov does not list all clinical trials that might be available.

GENE
BRCA2
ALTERATION
 T3085fs*26

RATIONALE
 BRCA2 loss or inactivating alterations may predict sensitivity to PARP inhibitors or to ATR inhibitors. The genomic finding observed here

may serve as inclusion criteria for 1 or more clinical trials listed below.

NCT05797168
PHASE 1/2

Phase I/IIa Study for AZD5335 as Monotherapy and in Combination With Anti-cancer Agents in Participants With Solid Tumors

TARGETS
 PARP1

LOCATIONS: Istanbul (Turkey (Türkiye)), Ankara (Turkey (Türkiye)), Berlin (Germany), Leipzig (Germany), Regensburg (Germany), Hanover (Germany), Mannheim (Germany), Freiburg im Breisgau (Germany), Cambridge (United Kingdom), London (United Kingdom)

NCT04341181
PHASE 2

ProTarget - A Danish Nationwide Clinical Trial on Targeted Cancer Treatment Based on Genomic Profiling

TARGETS
 ALK, RET, PD-L1, VEGFRs, EGFR, PARP, FGFR1, FGFR2, FGFR3, ERBB2, MEK, BRAF, SMO

LOCATIONS: Copenhagen (Denmark)

NCT04817956
PHASE 2

Improving Public Cancer Care by Implementing Precision Medicine in Norway

TARGETS
 PD-L1, VEGFA, ERBB2, ALK, RET, PARP, SMO, TRKB, TRKC, ROS1, TRKA, MEK, BRAF, PI3K-alpha, FGFR1, FGFR2, FGFR3, MET, KIT, ABL

LOCATIONS: Fredrikstad (Norway), Oslo (Norway), Skien (Norway), Drammen (Norway), Hamar (Norway), Kristiansand (Norway), Trondheim (Norway), Stavanger (Norway), Bergen (Norway), Førde (Norway)

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CLINICAL TRIALS

NCT02925234
PHASE 2

The Drug Rediscovery Protocol (DRUP Trial)

TARGETS

EGFR, PARP, BRAF, ABL, KIT, MEK, ERBB2, SMO, VEGFRs, RET, PD-1, ERBB4, FGFR1, FGFR2, FGFR3, MET, ROS1, PD-L1, VEGFA, TRKB, ALK, TRKC, TRKA, FGFRs, PDGFRA, CDK4, CDK6, AXL, FLT3, CSF1R, CTLA-4, PI3K-alpha

LOCATIONS: Almelo (Netherlands), Groningen (Netherlands), Hogeveen (Netherlands), Deventer (Netherlands), Drachten (Netherlands), Zwolle (Netherlands), Venlo (Netherlands), Apeldoorn (Netherlands), Arnhem (Netherlands), Leeuwarden (Netherlands)

NCT04657068
PHASE 1/2

A Study of ART0380 for the Treatment of Advanced or Metastatic Solid Tumors

TARGETS

ATR

LOCATIONS: Paris (France), Villejuif (France), Marseille (France), London (United Kingdom), Manchester (United Kingdom), Girona (Spain), Glasgow (United Kingdom), Badalona (Spain), Sabadell (Spain), Barcelona (Spain)

NCT05691491
PHASE 1/2

Testing the Combination of the Anti-Cancer Drugs Temozolomide and M1774 to Evaluate Their Safety and Effectiveness

TARGETS

ATR

LOCATIONS: Connecticut, Maryland, Pennsylvania, Tennessee, Missouri, Kansas

NCT05327010
PHASE 2

Testing the Combination of the Anti-cancer Drugs ZEN003694 (ZEN-3694) and Talazoparib in Patients With Advanced Solid Tumors, The CombET Trial

TARGETS

BRD4, BRDT, BRD2, BRD3, PARP

LOCATIONS: Maryland, Pennsylvania, Virginia, Illinois, North Carolina, Kentucky, Georgia, Missouri

NCT02693535
PHASE 2

TAPUR: Testing the Use of Food and Drug Administration (FDA) Approved Drugs That Target a Specific Abnormality in a Tumor Gene in People With Advanced Stage Cancer

TARGETS

CDK4, CDK6, FLT3, VEGFRs, CSF1R, KIT, RET, mTOR, ERBB2, MEK, BRAF, PARP, PD-1, CTLA-4, PD-L1, TRKB, ALK, TRKC, ROS1, TRKA, FGFRs

LOCATIONS: Maine

NCT06074692
PHASE 2

PARP Inhibition, Stereotactic Body Radiotherapy and Immunotherapy for Metastatic or Advanced Sarcoma (PRIMA): a Prospective Phase 2 Trial

TARGETS

PARP, PD-1

LOCATIONS: Shanghai (China)

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CLINICAL TRIALS

NCT05035745	PHASE 1/2
Selinexor & Talazoparib in Advanced Refractory Solid Tumors; Advanced/Metastatic Triple Negative Breast Cancer (START)	TARGETS XPO1, PARP
LOCATIONS: Singapore (Singapore)	

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CLINICAL TRIALS
GENE
CDKN2A
ALTERATION
 rearrangement intron 2

RATIONALE
 CDKN2A alterations may predict sensitivity to CDK4/6 inhibitors such as palbociclib, ribociclib, and abemaciclib. The genomic finding observed

here may serve as inclusion criteria for 1 or more clinical trials listed below.

NCT02925234
PHASE 2

The Drug Rediscovery Protocol (DRUP Trial)

TARGETS

EGFR, PARP, BRAF, ABL, KIT, MEK, ERBB2, SMO, VEGFRs, RET, PD-1, ERBB4, FGFR1, FGFR2, FGFR3, MET, ROS1, PD-L1, VEGFA, TRKB, ALK, TRKC, TRKA, FGFRs, PDGFRA, CDK4, CDK6, AXL, FLT3, CSF1R, CTLA-4, PI3K-alpha

LOCATIONS: Almelo (Netherlands), Groningen (Netherlands), Hoogeveen (Netherlands), Deventer (Netherlands), Drachten (Netherlands), Zwolle (Netherlands), Venlo (Netherlands), Apeldoorn (Netherlands), Arnhem (Netherlands), Leeuwarden (Netherlands)

NCT02693535
PHASE 2

TAPUR: Testing the Use of Food and Drug Administration (FDA) Approved Drugs That Target a Specific Abnormality in a Tumor Gene in People With Advanced Stage Cancer

TARGETS

CDK4, CDK6, FLT3, VEGFRs, CSF1R, KIT, RET, mTOR, ERBB2, MEK, BRAF, PARP, PD-1, CTLA-4, PD-L1, TRKB, ALK, TRKC, ROS1, TRKA, FGFRs

LOCATIONS: Maine

NCT04282031
PHASE 1/2

A Study of BPI-1178 in Patients With Advanced Solid Tumor and HR+/HER2- Breast Cancer

TARGETS

CDK6, CDK4, ER, Aromatase

LOCATIONS: Shanghai (China)

NCT05953350
PHASE 1/2

A Phase Ib/II Study Confirmed Inhibition of Autophagy Synergizes Anti-tumor Effect of High Dose CDK4/6i

TARGETS

CDK4, CDK6

LOCATIONS: Guangzhou (China)

NCT03454035
PHASE 1

Ulixertinib/Palbociclib in Patients With Advanced Pancreatic and Other Solid Tumors

TARGETS

CDK4, CDK6, ERK2, ERK1

LOCATIONS: North Carolina

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CLINICAL TRIALS

NCT04693468

PHASE 1

Talazoparib and Palbociclib, Axitinib, or Crizotinib for the Treatment of Advanced or Metastatic Solid Tumors, TalaCom Trial

TARGETS
PARP, CDK4, CDK6, VEGFRs, ALK, ROS1, AXL, TRKA, MET, TRKC

LOCATIONS: Texas

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CLINICAL TRIALS
GENE
KIT
ALTERATION

exon 11 deletion (Q556_V559>H), exon 13 (V654A), exon 18 (A829P)

RATIONALE

Activating mutations in KIT may confer sensitivity to Type-1 inhibitors of KIT such as bezugclastinib, based on clinical and pre-clinical evidence of benefit for patients with activating mutations in KIT exon 11- and KIT exon 17-mutated GIST. KIT amplification or activating mutations may predict sensitivity to small-molecule TKIs. Also, because KIT activation leads to activation of the PI3K-AKT-mTOR pathway, PI3K and mTOR pathway inhibitors may be relevant in a tumor with KIT activation. The

genomic finding observed here may serve as inclusion criteria for 1 or more clinical trials listed below. Mutation of residues within or adjacent to the KIT activation loop, as observed here, have demonstrated resistance to imatinib and sunitinib in patients with GIST. Mutations in KIT exons 13 and 14 may be associated with reduced sensitivity to ripretinib. The KIT V654A mutation is associated with resistance to imatinib and avapritinib and decreased sensitivity to regorafenib, ponatinib, and ripretinib.

NCT05957367
PHASE 1/2

A Study of DCC-3116 in Combination With Anticancer Therapies in Participants With Advanced Malignancies

TARGETS

KIT, PDGFRA, ULK1, ULK2, MEK, BRAF, EGFR

LOCATIONS: Frankfurt (Germany), Hamburg (Germany), Aarhus (Denmark), Florence (Italy), Milan (Italy), Nijmegen (Netherlands), Bern (Switzerland), Genova (Italy), Amsterdam (Netherlands), Paris (France)

NCT05159245
PHASE 2

The Finnish National Study to Facilitate Patient Access to Targeted Anti-cancer Drugs

TARGETS

BRAF, VEGFRs, RET, KIT, ERBB2, TRKB, ALK, TRKC, ROS1, TRKA, SMO, PD-L1, MEK, CDK4, CDK6

LOCATIONS: Helsinki (Finland), Turku (Finland), Tampere (Finland), Kuopio (Finland), Oulu (Finland)

NCT05443087
PHASE NULL

TARGETed Therapy Drug MONITORing in DIGestive Oncology

TARGETS

BRAF, VEGFRs, RET, KIT, mTOR, FLT3, CSF1R, MET, ROS1, MEK, EGFR

LOCATIONS: Strasbourg (France), Nancy (France), Dijon (France), Reims (France), Lille (France), Auxerre (France), Amiens (France), Paris (France), Villejuif (France), Clichy (France)

NCT04116541
PHASE 2

A Study Evaluating the Activity of Anti-cancer Treatments Targeting Tumor Molecular Alterations/ Characteristics in Advanced / Metastatic Tumors.

TARGETS

CDK6, CDK4, MDM2, MET, ROS1, RET, VEGFRs, ALK, BRAF, KIT, MEK, PDGFRA

LOCATIONS: Strasbourg (France), Lyon (France), Marseille (France), Nice (France), Villejuif (France), Caen (France), Toulouse (France), Bordeaux (France)

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CLINICAL TRIALS

NCT02925234
PHASE 2

The Drug Rediscovery Protocol (DRUP Trial)

TARGETS
EGFR, PARP, BRAF, ABL, KIT, MEK, ERBB2, SMO, VEGFRs, RET, PD-1, ERBB4, FGFR1, FGFR2, FGFR3, MET, ROS1, PD-L1, VEGFA, TRKB, ALK, TRKC, TRKA, FGFRs, PDGFRA, CDK4, CDK6, AXL, FLT3, CSF1R, CTLA-4, PI3K-alpha

LOCATIONS: Almelo (Netherlands), Groningen (Netherlands), Hogeveen (Netherlands), Deventer (Netherlands), Drachten (Netherlands), Zwolle (Netherlands), Venlo (Netherlands), Apeldoorn (Netherlands), Arnhem (Netherlands), Leeuwarden (Netherlands)

NCT05024214
PHASE 1/2

Phase Ib/II Trial of Envafohimab Plus Lenvatinib for Subjects With Solid Tumors

TARGETS
PD-L1, FGFRs, RET, PDGFRA, VEGFRs, KIT, FLT3, CSF1R

LOCATIONS: Ürümqi (China), Taiyuan (China), Beijing (China), Tianjin (China), Kunming (China), Zhengzhou (China), Jinan (China), Shenyang (China), Dalian (China), Bengbu (China)

NCT04199026
PHASE NULL

Implantable Microdevice for the Delivery of Drugs and Their Effect on Tumors in Patients With Metastatic or Recurrent Sarcoma

TARGETS
TOP1, mTOR, IGF1R, FGFR3, KIT, FGFR1, VEGFRs, FGFR2

LOCATIONS: Texas

NCT07379047
PHASE 2/3

Efficacy and Safety of NB003 in Patients With Advanced Gastrointestinal Stromal Tumors (GIST)

TARGETS
KIT, PDGFRA, BRAF, VEGFRs, RET

LOCATIONS: Beijing (China), Qingdao (China), Wuhan (China), Hefei (China)

NCT02693535
PHASE 2

TAPUR: Testing the Use of Food and Drug Administration (FDA) Approved Drugs That Target a Specific Abnormality in a Tumor Gene in People With Advanced Stage Cancer

TARGETS
CDK4, CDK6, FLT3, VEGFRs, CSF1R, KIT, RET, mTOR, ERBB2, MEK, BRAF, PARP, PD-1, CTLA-4, PD-L1, TRKB, ALK, TRKC, ROS1, TRKA, FGFRs

LOCATIONS: Maine

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CLINICAL TRIALS

NCT06081907

PHASE 1/2

The Efficacy and Safety of IBI363 in Solid Tumors.

TARGETS
IL-2R, PD-1, CD73, FGFRs, RET,
PDGFRA, VEGFRs, KIT

LOCATIONS: Changsha (China)

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CLINICAL TRIALS
GENE
MTOR
ALTERATION
A1459D

RATIONALE
MTOR mutation may lead to activation of the mTOR pathway, and may therefore indicate sensitivity to mTOR inhibitors. The genomic

finding observed here may serve as inclusion criteria for 1 or more clinical trials listed below.

NCT04199026
PHASE NULL

Implantable Microdevice for the Delivery of Drugs and Their Effect on Tumors in Patients With Metastatic or Recurrent Sarcoma

TARGETS
TOP1, mTOR, IGF1R, FGFR3, KIT, FGFR1, VEGFRs, FGFR2

LOCATIONS: Texas

NCT07109726
PHASE 1/2

A Phase 1/2 Trial of TER-2013 in Patients With Solid Tumors Harboring AKT/PI3K/PTEN Pathway Alterations

TARGETS
AKT1, ER

LOCATIONS: Massachusetts, Virginia, Ohio, Wisconsin, Illinois, Minnesota, North Carolina, Missouri, Nebraska, Tennessee

NCT02693535
PHASE 2

TAPUR: Testing the Use of Food and Drug Administration (FDA) Approved Drugs That Target a Specific Abnormality in a Tumor Gene in People With Advanced Stage Cancer

TARGETS
CDK4, CDK6, FLT3, VEGFRs, CSF1R, KIT, RET, mTOR, ERBB2, MEK, BRAF, PARP, PD-1, CTLA-4, PD-L1, TRKB, ALK, TRKC, ROS1, TRKA, FGFRs

LOCATIONS: Maine

NCT05036226
PHASE 1/2

COAST Therapy in Advanced Solid Tumors and Prostate Cancer

TARGETS
DDR2, ABL, SRC, KIT, mTOR

LOCATIONS: South Carolina

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APPENDIX
Variants of Unknown Significance

NOTE One or more variants of unknown significance (VUS) were detected in this patient's tumor. These variants may not have been adequately characterized in the scientific literature at the time this report was issued, and/or the genomic context of these alterations makes their significance unclear. We choose to include them here in the event that they become clinically meaningful in the future.

AR NM_000044.2: c.2668G>C (p.V890L) chrX:66943588 0.21% VAF	CDKN2C NM_001262.2: c.204_211delinsCA (p.T69_F71delinsI) chr1:51439639-51439646 5.8% VAF	ERBB4 NM_005235.2: c.284G>A (p.R95H) chr2:212812292 48.4% VAF	IGF1R NM_000875.3: c.4082T>G (p.L1361R) chr15:99500649 51.1% VAF
MLH1 NM_000249.3: c.25C>T (p.R9W) chr3:37035063 0.35% VAF	ZNF703 NM_025069.1: c.1204_1205insN[42] (p.A401_H402insX[14]) chr8:37555584 42.6% VAF		

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APPENDIX

Genes assayed in FoundationOne®Liquid CDx

FoundationOne Liquid CDx interrogates 324 genes, including 309 genes with complete exonic (coding) coverage and 15 genes with only select non-coding coverage (indicated with an *); 75 genes (indicated in bold) are captured with increased sensitivity and have complete exonic (coding) coverage unless otherwise noted.

ABL1 Exons 4-9	ACVR1B	AKT1 Exon 3	AKT2	AKT3	ALK Exons 20-29, Introns 18, 19	ALOX12B	AMER1 (FAM123B or WTX)	APC
AR	ARAF Exons 4, 5, 7, 11, 13, 15, 16	ARFRP1	ARID1A	ASXL1	ATM	ATR	ATRX	AURKA
AURKB	AXIN1	AXL	BAP1	BARD1	BCL2	BCL2L1	BCL2L2	BCL6
BCOR	BCORL1	BCR* Introns 8, 13, 14	BRAF Exons 11-18, Introns 7-10	BRCA1 Introns 2, 7, 8, 12, 16, 19, 20	BRCA2 Intron 2	BRD4	BRIP1	BTG1
BTG2	BTK Exons 2, 15	CALR	CARD11	CASP8	CBFB	CBL	CCND1	CCND2
CCND3	CCNE1	CD22	CD70	CD74* Introns 6-8	CD79A	CD79B	CD274 (PD-L1)	CDC73
CDH1	CDK12	CDK4	CDK6	CDK8	CDKN1A	CDKN1B	CDKN2A	CDKN2B
CDKN2C	CEBPA	CHEK1	CHEK2	CIC	CREBBP	CRKL	CSF1R	CSF3R
CTCF	CTNNA1	CTNNB1 Exon 3	CUL3	CUL4A	CXCR4	CYP17A1	DAXX	DDR1
DDR2 Exons 5, 17, 18	DIS3	DNMT3A	DOT1L	EED	EGFR Introns 7, 15, 24-27	EMSY (C11orf30)	EP300	EPHA3
EPHB1	EPHB4	ERBB2	ERBB3 Exons 3, 6, 7, 8, 10, 12, 20, 21, 23, 24, 25	ERBB4	ERCC4	ERG	ERRFI1	ESR1 Exons 4-8
ETV4* Intron 8	ETV5* Introns 6, 7	ETV6* Introns 5, 6	EWSR1* Introns 7-13	EZH2 Exons 4, 16, 17, 18	EZR* Introns 9-11	FANCA	FANCC	FANCG
FANCL	FAS	FBXW7	FGF10	FGF12	FGF14	FGF19	FGF23	FGF3
FGF4	FGF6	FGFR1 Introns 1, 5, Intron 17	FGFR2 Intron 1, Intron 17	FGFR3 Exons 7, 9 (alternative designation exon 10), 14, 18, Intron 17	FGFR4	FH	FLCN	FLT1
FLT3 Exons 14, 15, 20	FOXL2	FUBP1	GABRA6	GATA3	GATA4	GATA6	GID4 (C17orf39)	GNAI1 Exons 4, 5
GNA13	GNAQ Exons 4, 5	GNAS Exons 1, 8	GRM3	GSK3B	H3-3A (H3F3A)	HDAC1	HGF	HNF1A
HRAS Exons 2, 3	HSD3B1	ID3	IDH1 Exon 4	IDH2 Exon 4	IGF1R	IKBKE	IKZF1	INPP4B
IRF2	IRF4	IRS2	JAK1	JAK2 Exon 14	JAK3 Exons 5, 11, 12, 13, 15, 16	JUN	KDM5A	KDM5C
KDM6A	KDR	KEAP1	KEL	KIT Exons 8, 9, 11, 12, 13, 17, Intron 16	KLHL6	KMT2A (MLL) Introns 6, 8-11, Intron 7	KMT2D (MLL2)	KRAS

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Electronically signed by Tyler Janovitz, MD, PhD | 30 August 2026
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Genes assayed in FoundationOne®Liquid CDx

FoundationOne Liquid CDx interrogates 324 genes, including 309 genes with complete exonic (coding) coverage and 15 genes with only select non-coding coverage (indicated with an *); 75 genes (indicated in bold) are captured with increased sensitivity and have complete exonic (coding) coverage unless otherwise noted.

<i>LTK</i>	<i>LYN</i>	<i>MAF</i>	MAP2K1 (MEK1) Exons 2, 3	MAP2K2 (MEK2) Exons 2-4, 6, 7	<i>MAP2K4</i>	<i>MAP3K1</i>	<i>MAP3K13</i>	<i>MAPK1</i>
<i>MCL1</i>	MDM2	<i>MDM4</i>	<i>MED12</i>	<i>MEF2B</i>	<i>MEN1</i>	<i>MERTK</i>	MET	<i>MITF</i>
<i>MKNK1</i>	<i>MLH1</i>	MPL Exon 10	<i>MRE11</i> (MRE11A)	<i>MSH2</i> Intron 5	<i>MSH3</i>	<i>MSH6</i>	<i>MST1R</i>	<i>MTAP</i>
MTOR Exons 19, 30, 39, 40, 43-45, 47, 48, 53, 56	<i>MUTYH</i>	<i>MYB*</i> Intron 14	MYC Intron 1	<i>MYCL</i> (MYCL1)	MYCN	MYD88 Exon 4	<i>NBN</i>	NF1
<i>NF2</i>	<i>NFE2L2</i>	<i>NFKBIA</i>	<i>NKX2-1</i>	<i>NOTCH1</i>	<i>NOTCH2</i> Intron 26	<i>NOTCH3</i>	NPM1 Exons 4-6, 8, 10	NRAS Exons 2, 3
<i>NSD2</i> (WHSC1 or MMSET)	<i>NSD3</i> (WHSC1L1)	<i>NT5C2</i>	NTRK1 Exons 14, 15, Introns 8-11	<i>NTRK2</i> Intron 12	NTRK3 Exons 16, 17	<i>NUTM1*</i> Intron 1	<i>P2RY8</i>	PALB2
<i>PARP1</i>	<i>PARP2</i>	<i>PARP3</i>	<i>PAX5</i>	<i>PBRM1</i>	<i>PDCD1</i> (PD-1)	PDCD1LG2 (PD-L2)	PDGFRA Exons 12, 18, Introns 7, 9, 11	PDGFRB Exons 12-21, 23
<i>PDK1</i>	<i>PIK3C2B</i>	<i>PIK3C2G</i>	PIK3CA Exons 2, 3, 5-8, 10, 14, 19, 21 (Coding Exons 1, 2, 4-7, 9, 13, 18, 20)	<i>PIK3CB</i>	<i>PIK3R1</i>	<i>PIM1</i>	<i>PMS2</i>	<i>POLD1</i>
<i>POLE</i>	<i>PPARG</i>	<i>PPP2R1A</i>	<i>PPP2R2A</i>	<i>PRDM1</i>	<i>PRKAR1A</i>	<i>PRKCI</i>	<i>PRKN</i> (PARK2)	<i>PTCH1</i>
PTEN	PTPN11	<i>PTPRO</i>	<i>QKI</i>	<i>RAC1</i>	<i>RAD21</i>	<i>RAD51</i>	<i>RAD51B</i>	<i>RAD51C</i>
<i>RAD51D</i>	<i>RAD52</i>	<i>RAD54L</i>	RAF1 Exons 3, 4, 6, 7, 10, 14, 15, 17, Introns 4-8	<i>RARA</i> Intron 2	RB1	<i>RBM10</i>	<i>REL</i>	RET Introns 7, 8, Exons 11, 13-16, Introns 9-11
<i>RICTOR</i>	<i>RNF43</i>	ROS1 Exons 31, 36-38, 40, Introns 31-35	<i>RPTOR</i>	<i>RSP02*</i> Intron 1	<i>SDC4*</i> Intron 2	<i>SDHA</i>	<i>SDHB</i>	<i>SDHC</i>
<i>SDHD</i>	<i>SETD2</i>	<i>SF3B1</i>	<i>SGK1</i>	<i>SLC34A2*</i> Intron 4	<i>SMAD2</i>	<i>SMAD4</i>	<i>SMARCA4</i>	<i>SMARCB1</i>
SMO	<i>SNCAIP</i>	<i>SOCs1</i>	<i>SOX2</i>	<i>SOX9</i>	<i>SPEN</i>	<i>SPOP</i>	<i>SRC</i>	<i>STAG2</i>
<i>STAT3</i>	STK11	<i>SUFU</i>	<i>SYK</i>	<i>TBX3</i>	<i>TEK</i>	<i>TENT5C</i> (FAM46C)	<i>TERC*</i> ncRNA	TERT* Promoter
<i>TET2</i>	<i>TGFBR2</i>	<i>TIPARP</i>	<i>TMPRSS2*</i> Introns 1-3	<i>TNFAIP3</i>	<i>TNFRSF14</i>	TP53	<i>TSC1</i>	<i>TSC2</i>
<i>TYRO3</i>	<i>U2AF1</i>	VEGFA	<i>VHL</i>	<i>WT1</i>	<i>XPO1</i>	<i>XRCC2</i>	<i>ZNF217</i>	<i>ZNF703</i>

ADDITIONAL ASSAYS: FOR THE DETECTION OF SELECT CANCER BIOMARKERS

Microsatellite (MS) status
Blood Tumor Mutational Burden (bTMB)
ctDNA Tumor Fraction

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APPENDIX
About FoundationOne® Liquid CDx

FoundationOne Liquid CDx fulfills the requirements of the European Directive 98/79 EC for in vitro diagnostic medical devices and is registered as a CE-IVD product by Foundation Medicine's EU Authorized Representative, Qarad b.v.b.a, Cipalstraat 3, 2440 Geel, Belgium. The CE-IVD regulatory status of FoundationOne Liquid CDx is applicable in countries that accept and/or recognize the CE mark.

Foundation Medicine GmbH is accredited by DAkkS according to DIN EN ISO 15189:2013. The accreditation only applies to the scope of accreditation listed in certificate D-ML-21105-01-00.


ABOUT FOUNDATIONONE LIQUID CDx

FoundationOne Liquid CDx was developed and its performance characteristics determined by Foundation Medicine, Inc. (Foundation Medicine). FoundationOne Liquid CDx may be used for clinical purposes and should not be regarded as purely investigational or for research only. Foundation Medicine's clinical reference laboratories are qualified to perform high-complexity clinical testing.

Please refer to technical information for performance specification details.

INTENDED USE

FoundationOne Liquid CDx is a next generation sequencing based *in vitro* diagnostic device that analyzes 324 genes. Substitutions and insertion and deletion alterations (indels) are reported in 311 genes, copy number alterations (CNAs) are reported in 310 genes, and gene rearrangements are reported in 324 genes. The test also detects the genomic signatures blood tumor mutational burden (bTMB), microsatellite instability (MSI), and ctDNA tumor fraction. FoundationOne Liquid CDx utilizes circulating cell-free DNA (cfDNA) isolated from plasma derived from the anti-coagulated peripheral whole blood of cancer patients. The test is intended to be used as a companion diagnostic to identify patients who may benefit from treatment with targeted therapies in accordance with the approved therapeutic product labeling. Additionally, FoundationOne Liquid CDx is intended to provide tumor mutation profiling to be used by qualified health care professionals in accordance with professional guidelines in oncology for patients with malignant neoplasms.

TEST PRINCIPLES

The FoundationOne Liquid CDx assay is performed

exclusively as a laboratory service using circulating cell-free DNA (cfDNA) isolated from plasma derived from anti-coagulated peripheral whole blood from patients with solid malignant neoplasms. The assay employs a single DNA extraction method to obtain cfDNA from plasma from whole blood. Extracted cfDNA undergoes whole-genome shotgun library construction and hybridization-based capture of 324 cancer-related genes including coding exons and select introns of 309 genes, as well as only select intronic regions or non-coding regions of 15 genes. Hybrid-capture selected libraries are sequenced with deep coverage using the NovaSeq® 6000 platform. Sequence data are processed using a customized analysis pipeline designed to accurately detect genomic alterations, including base substitutions, indels, select copy number variants, and select genomic rearrangements. Substitutions and insertion and deletion alterations (indels) are reported in 311 genes, copy number alterations (CNAs) are reported in 310 genes, and gene rearrangements are reported in 324 genes. The assay also reports ctDNA tumor fraction, and genomic signatures including MSI and bTMB. A subset of targeted regions in 75 genes is baited for increased sensitivity.

THE REPORT

Incorporates analyses of peer-reviewed studies and other publicly available information identified by Foundation Medicine; these analyses and information may include associations between a molecular alteration (or lack of alteration) and one or more drugs with potential clinical benefit (or potential lack of clinical benefit), including drug candidates that are being studied in clinical research. *Note:* A finding of biomarker alteration does not necessarily indicate pharmacologic effectiveness (or lack thereof) of any drug or treatment regimen; a finding of no biomarker alteration does not necessarily indicate lack of pharmacologic effectiveness (or effectiveness) of any drug or treatment regimen.

DISEASE AND TUMOR TYPE

The 'DISEASE' and 'TUMOR TYPE' fields in this report represent a standardized term mapped from the diagnosis submitted by the ordering physician and may be adjusted at the discretion of a Foundation Medicine pathologist. This service is not accredited under ISO15189.

QUALIFIED ALTERATION CALLS (EQUIVOCAL)

All equivocal calls, regardless of alteration type, imply that there is adequate evidence to call the alteration with confidence. However, the

repeatability of equivocal calls may be lower than non-equivocal calls.

COPY NUMBER LOSS CALLS

The FoundationOne Liquid CDx assay detects copy number loss in the following genes: *BRCA1*, *BRCA2*, and *PTEN*.

RANKING OF THERAPIES AND CLINICAL TRIALS

Ranking of Therapies in Summary Table

Therapies are ranked based on the following criteria: Therapies with clinical benefit (ranked alphabetically within each evidence category), followed by therapies associated with resistance (when applicable).

Ranking of Clinical Trials

Pediatric trial qualification → Geographical proximity → Later trial phase.

LIMITATIONS

1. For *in vitro* diagnostic use.
2. For prescription use only. This test must be ordered by a qualified medical professional in accordance with clinical laboratory regulations.
3. A negative result does not rule out the presence of a mutation below the limits of detection of the assay. Patients for whom no companion diagnostic alterations are detected should be considered for confirmation with an appropriately validated tumor tissue test, if available.
4. The FoundationOne Liquid CDx assay does not detect heterozygous deletions.
5. The test is not intended to provide information on cancer predisposition.
6. Performance has not been validated for cfDNA input below the specified minimum input.
7. Tissue TMB and blood TMB (bTMB) are estimated from the number of synonymous and nonsynonymous single-nucleotide variants (SNVs) and insertions and deletions (indels) per area of coding genome sampled, after the removal of known and likely oncogenic driver events and germline SNPs. Tissue TMB is calculated based on variants with an allele frequency of ≥5%, and bTMB is calculated based on variants with an allele frequency of ≥0.5%.
8. ctDNA tumor fraction is the percentage of circulating tumor DNA (ctDNA) present in a cell-free DNA (cfDNA) sample. The ctDNA tumor fraction estimate integrates multiple distinct genomic features, including modeled aneuploidy and the observed allele frequencies of somatic short variants and rearrangements.

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9. Microsatellite instability (MSI) is a condition of genetic hypermutability that generates excessive amounts of short insertion/deletion mutations in the tumor genome; it generally occurs at microsatellite DNA sequences and is caused by a deficiency in DNA mismatch repair (MMR) in the tumor. The MSI algorithm is based on genome wide analysis of 1765 microsatellite loci and not based on the 5 or 7 MSI loci described in current clinical practice guidelines for solid tissue testing.
10. Genomic findings from circulating cell-free DNA (cfDNA) may originate from circulating tumor DNA fragments, germline alterations, or non-tumor somatic alterations, such as clonal hematopoiesis of indeterminate potential (CHIP). Genes with alterations that may be derived from CHIP include, but are not limited to: *ASXL1*, *ATM*, *CBL*, *CHEK2*, *DNMT3A*, *JAK2*, *KMT2D* (*MLL2*), *MPL*, *MYD88*, *SF3B1*, *TET2*, *TP53*, and *U2AF1*.
11. Alterations reported may include somatic (not inherited) or germline (inherited) alterations; however, the test does not distinguish between germline and somatic alterations. If a reported alteration is suspected to be germline, confirmatory testing should be considered in the appropriate clinical context.
12. The test is not intended to replace germline testing or to provide information about cancer predisposition.

REPORT HIGHLIGHTS

The Report Highlights includes select genomic and therapeutic information with potential impact on patient care and treatment that is specific to the genomics and tumor type of the sample analyzed. This section may highlight information including targeted therapies with potential sensitivity or resistance; evidence-matched clinical trials; and variants with potential diagnostic, prognostic, nontargeted treatment, germline, or clonal hematopoiesis implications. Information included in the Report Highlights is expected to evolve with advances in scientific and clinical research. Findings included in the Report Highlights should be considered in the context of all other information in this report and other relevant patient information. Decisions on patient care and treatment are the responsibility of the treating physician.

VARIANTS TO CONSIDER FOR FOLLOW-UP GERMLINE TESTING

The variants indicated for consideration of follow-up germline testing are 1) limited to reportable short variants with a protein effect listed in the

ClinVar genomic database (Landrum et al., 2018; 29165669) as Pathogenic, Pathogenic/Likely Pathogenic, or Likely Pathogenic (by an expert panel or multiple submitters), 2) associated with hereditary cancer-predisposing disorder(s), 3) detected at an allele frequency of >30%, and 4) in select genes informed by the ESMO Precision Medicine Working Group (Kuzbari et al., 2023; 36529447) and ASCO Guidelines (Tung et al., 2024; 38759122). The selected genes are *ATM*, *BAP1*, *BARD1*, *BRCA1*, *BRCA2*, *BRIP1*, *CHEK2*, *FH*, *FLCN*, *MLH1*, *MSH2*, *MSH6*, *MUTYH*, *PALB2*, *PMS2*, *POLD1*, *POLE*, *PTCH1*, *RAD51C*, *RAD51D*, *RET*, *SDHA*, *SDHB*, *SDHC*, *SDHD*, *SMARCB1*, *SUFU*, *TSC2*, and *VHL*, and are not inclusive of all cancer susceptibility genes. The content in this report should not substitute for genetic counseling or follow-up germline testing, which is needed to distinguish whether a finding in this patient's tumor sequencing is germline or somatic. Interpretation should be based on clinical context.

VARIANTS THAT MAY REPRESENT CLONAL HEMATOPOIESIS

Variants that may represent clonal hematopoiesis (CH) are limited to select reportable short variants in defined genes identified in solid tumors only. Variant selection was determined based on gene tumor-suppressor or oncogene status, known role in solid tumors versus hematological malignancies, and literature prevalence. The defined genes are *ASXL1*, *ATM*, *CBL*, *CHEK2*, *DNMT3A*, *IDH2*, *JAK2*, *KMT2D* (*MLL2*), *MPL*, *MYD88*, *SF3B1*, *TET2*, and *U2AF1* and are not inclusive of all CH genes. The content in this report should not substitute for dedicated hematological workup. Comprehensive genomic profiling of solid tumors detects nontumor alterations that are due to CH. Patient-matched peripheral blood mononuclear cell sequencing is required to conclusively determine if this alteration is present in tumor or is secondary to CH. Interpretation should be based on clinical context.

NATIONAL COMPREHENSIVE CANCER NETWORK® (NCCN®) CATEGORIZATION

Biomarker and genomic findings detected may be associated with certain entries within the NCCN Drugs & Biologics Compendium® (NCCN Compendium®) (www.nccn.org). The NCCN Categories of Evidence and Consensus indicated reflect the highest possible category for a given therapy in association with each biomarker or genomic finding. Please note, however, that the accuracy and applicability of these NCCN categories within a report may be impacted by the patient's clinical history, additional biomarker information, age, and/or co-occurring alterations. For additional information on the NCCN categories, please refer to

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LEVEL OF EVIDENCE NOT PROVIDED

Drugs with potential clinical benefit (or potential lack of clinical benefit) are not evaluated for source or level of published evidence.

NO GUARANTEE OF CLINICAL BENEFIT

This report makes no promises or guarantees that a particular drug will be effective in the treatment of disease in any patient. This report also makes no promises or guarantees that a drug with potential lack of clinical benefit will in fact provide no clinical benefit.

NO GUARANTEE OF REIMBURSEMENT

Foundation Medicine makes no promises or guarantees that a healthcare provider, insurer or other third party payor, whether private or governmental, will reimburse a patient for the cost of FoundationOne Liquid CDx.

TREATMENT DECISIONS ARE THE RESPONSIBILITY OF PHYSICIAN

Drugs referenced in this Report may not be suitable for a particular patient. The selection of any, all or none of the drugs associated with potential clinical benefit (or potential lack of clinical benefit) resides entirely within the discretion of the treating physician. Indeed, the information in this Report must be considered in conjunction with all other relevant information regarding a particular patient, before the patient's treating physician recommends a course of treatment. Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all applicable information concerning the patient's condition, such as patient and family history, physical examinations, information from other diagnostic tests, and patient preferences, in accordance with the standard of care in a given community. A treating physician's decisions should not be based on a single test, such as this test or the information contained in this report.

Certain sample of variant characteristics may result in reduced sensitivity. These include: low sample quality, deletions and insertions >40bp, or

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About FoundationOne®Liquid CDx

repetitive/high homology sequences.
FoundationOne Liquid CDx is performed using cell-free DNA, and as such germline events may not be reported.

SELECT ABBREVIATIONS

ABBREVIATION	DEFINITION
CR	Complete response
ctDNA	Circulating tumor DNA
DCR	Disease control rate
DFS	Disease-free survival
DOR	Duration of response
EFS	Event-free survival
ER	Estrogen receptor
HR +/-	Hormone-receptor positive/negative
ITD	Internal tandem duplication
MR	Molecular response
MMR	Mismatch repair
Muts/Mb	Mutations per megabase
NOS	Not otherwise specified
ORR	Objective response rate
OS	Overall survival
mOS	Median overall survival
PD	Progressive disease
PFS	Progression-free survival
mPFS	Median progression-free survival
PR	Partial response
PSA	Prostate-specific antigen
R/R	Relapsed or refractory
SD	Stable disease
TKI	Tyrosine kinase inhibitor
CRC	Colorectal cancer
HCC	Hepatocellular carcinoma
HNSCC	Head and neck squamous cell carcinoma
NSCLC	Non-small cell lung cancer
RCC	Renal cell carcinoma
SCC	Squamous cell carcinoma

REFERENCE SEQUENCE INFORMATION

Sequence data is mapped to the human genome, Genome Reference Consortium Human Build 37 (GRCh37), also known as hg19.

SOFTWARE VERSION INFORMATION

MR Suite Version (RG) 10.0.1
MR Reporting Config Version 91
Analysis Pipeline Version v3.47.0
Computational Biology Suite Version 7.5.1

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APPENDIX
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